

10 (short)

- (a) The work hardening contribution to the yield stress $\Delta\sigma_{wh}$ can be assumed to be proportional to $\sqrt{\rho_d}$, where ρ_d is the dislocation density. A hot rolled sheet of pure copper (Cu) has a yield stress of 90 MPa and $\rho_d = 10^{13} \text{ m}\cdot\text{m}^{-3}$. A cold rolled sheet of the same material has a yield stress of 360 MPa and $\rho_d = 10^{15} \text{ m}\cdot\text{m}^{-3}$. Find the yield stress of fully annealed pure Cu. [3]
- (b) The solid solution hardening contribution to the yield stress $\Delta\sigma_{ss}$ can be assumed to be proportional to $(C_{ss})^n$, where C_{ss} is the concentration in solid solution. An annealed Cu alloy containing $C_{ss} = 10 \%$ Ni has a yield stress of 110 MPa, while an alloy containing $C_{ss} = 20 \%$ Ni has a yield stress of 131 MPa. Find the exponent n . [3]
- (c) A Cu alloy containing 15 % Ni is cold rolled to a dislocation density $\rho_d = 10^{14} \text{ m}\cdot\text{m}^{-3}$, and has a yield stress of 216 MPa. Assess whether these values are consistent with the theory that the hardening contributions $\Delta\sigma_{wh}$ and $\Delta\sigma_{ss}$ are additive. [4]